

COGNIZANT

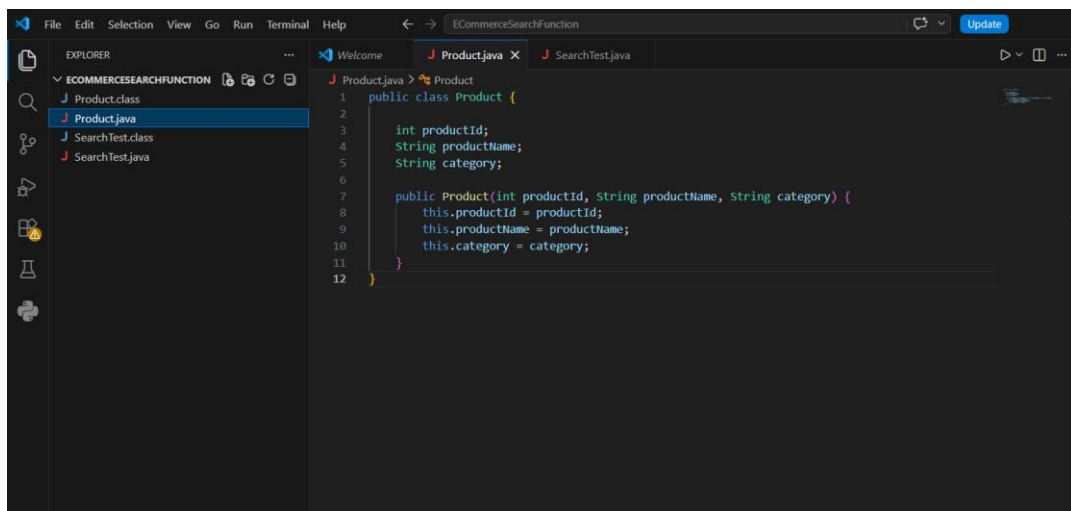
Mandatory Hands-on Exercises

-KHUSHI AHI

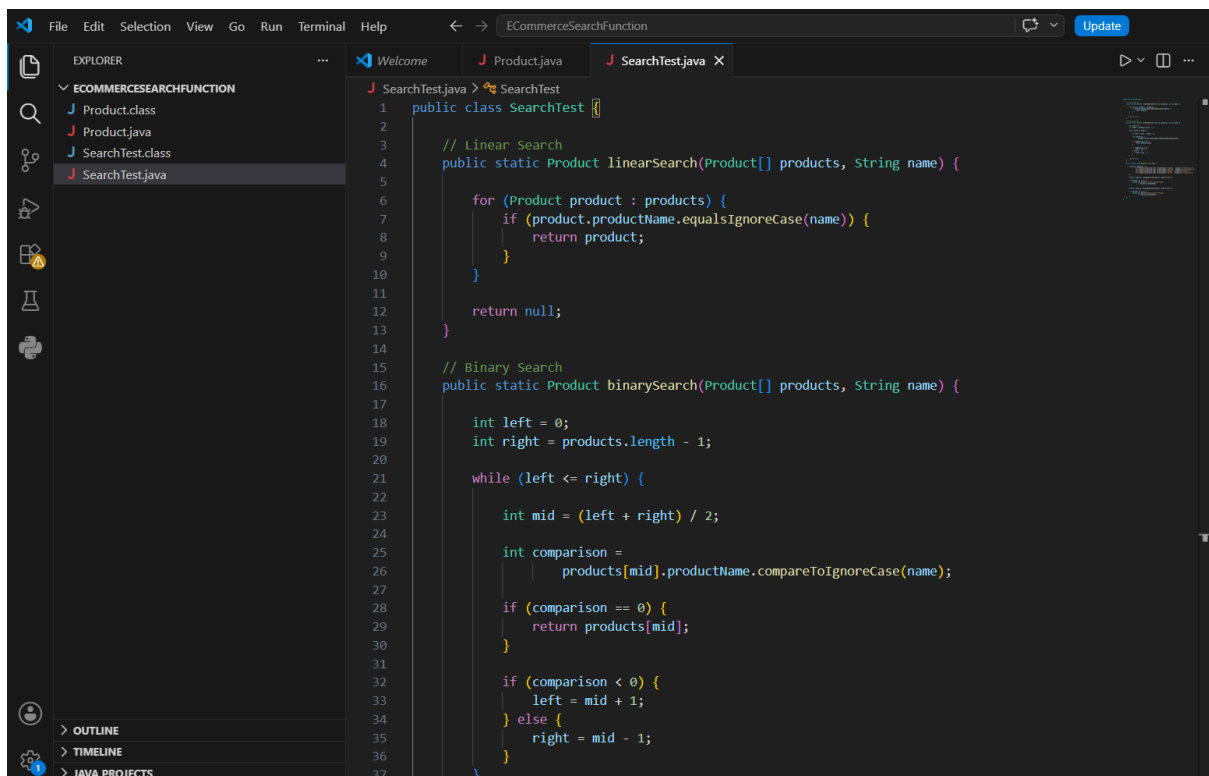
TOPIC: Data Structures and Algorithms

Exercise 2: E-commerce Platform Search Function

CODE



```
1 public class Product {
2
3     int productId;
4     String productName;
5     String category;
6
7     public Product(int productId, String productName, String category) {
8         this.productId = productId;
9         this.productName = productName;
10        this.category = category;
11    }
12 }
```



```
1 public class SearchTest {
2
3     // Linear Search
4     public static Product linearSearch(Product[] products, String name) {
5
6         for (Product product : products) {
7             if (product.productName.equalsIgnoreCase(name)) {
8                 return product;
9             }
10        }
11
12        return null;
13    }
14
15    // Binary Search
16    public static Product binarySearch(Product[] products, String name) {
17
18        int left = 0;
19        int right = products.length - 1;
20
21        while (left <= right) {
22
23            int mid = (left + right) / 2;
24
25            int comparison =
26                products[mid].productName.compareToIgnoreCase(name);
27
28            if (comparison == 0) {
29                return products[mid];
30            }
31
32            if (comparison < 0) {
33                left = mid + 1;
34            } else {
35                right = mid - 1;
36            }
37        }
38    }
39 }
```

```
38
39     return null;
40 }
41
Run | Debug
42 public static void main(String[] args) {
43
44     Product[] products = {
45         new Product(productId: 101, productName: "Laptop", category: "Electronics"),
46         new Product(productId: 102, productName: "Mobile", category: "Electronics"),
47         new Product(productId: 103, productName: "Shoes", category: "Fashion"),
48         new Product(productId: 104, productName: "Watch", category: "Accessories")
49     };
50
51     Product result1 = linearSearch(products, name: "Shoes");
52
53     if (result1 != null) {
54         System.out.println("Linear Search Found: "
55             + result1.productName);
56     }
57
58     Product result2 = binarySearch(products, name: "Shoes");
59
60     if (result2 != null) {
61         System.out.println("Binary Search Found: "
62             + result2.productName);
63     }
64 }
65 }
```

OUTPUT

```
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

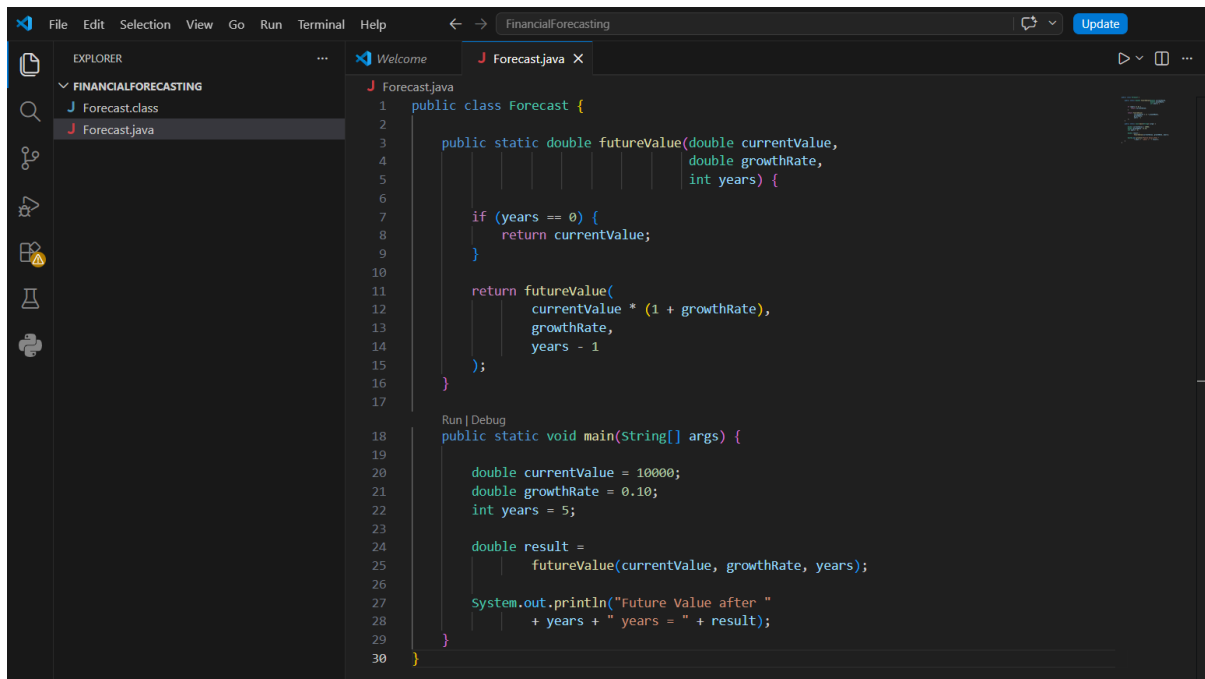
C:\Users\Khushi Ahi\OneDrive\Desktop\ECommerceSearchFunction>javac *.java

C:\Users\Khushi Ahi\OneDrive\Desktop\ECommerceSearchFunction>java SearchTest
Linear Search Found: Shoes
Binary Search Found: Shoes

C:\Users\Khushi Ahi\OneDrive\Desktop\ECommerceSearchFunction>
```

Exercise 7: Financial Forecasting

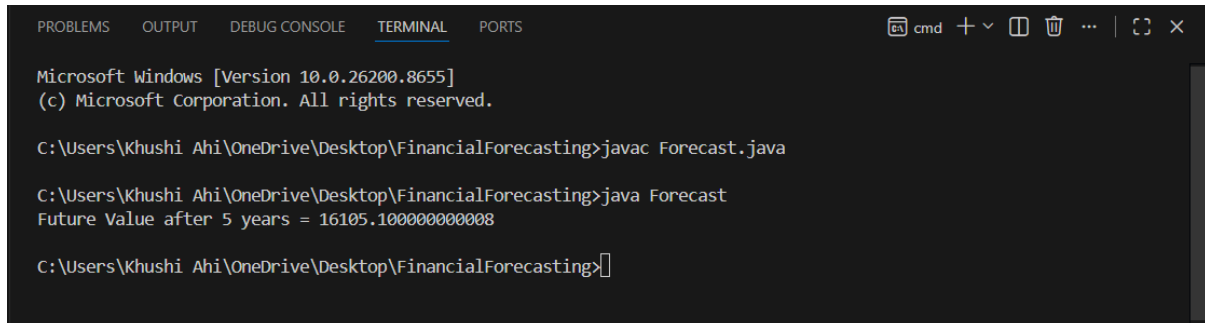
CODE



The screenshot shows an IDE window titled 'FinancialForecasting'. The Explorer pane on the left shows a project named 'FINANCIALFORECASTING' with two files: 'Forecast.class' and 'Forecast.java'. The main editor displays the code for 'Forecast.java'. The code defines a 'Forecast' class with a static method 'futureValue' and a 'main' method. The 'futureValue' method calculates the future value of an investment based on the current value, growth rate, and number of years. The 'main' method initializes the values and prints the result.

```
1 public class Forecast {
2
3     public static double futureValue(double currentValue,
4                                     double growthRate,
5                                     int years) {
6
7         if (years == 0) {
8             return currentValue;
9         }
10
11         return futureValue(
12             currentValue * (1 + growthRate),
13             growthRate,
14             years - 1
15         );
16     }
17
18     Run | Debug
19     public static void main(String[] args) {
20
21         double currentValue = 10000;
22         double growthRate = 0.10;
23         int years = 5;
24
25         double result =
26             futureValue(currentValue, growthRate, years);
27
28         System.out.println("Future Value after "
29                             + years + " years = " + result);
30     }
```

OUTPUT:



The screenshot shows a terminal window with the following output:

```
Microsoft Windows [Version 10.0.26200.8655]
(c) Microsoft Corporation. All rights reserved.

C:\Users\Khushi Ahi\OneDrive\Desktop\FinancialForecasting>javac Forecast.java

C:\Users\Khushi Ahi\OneDrive\Desktop\FinancialForecasting>java Forecast
Future Value after 5 years = 16105.100000000008

C:\Users\Khushi Ahi\OneDrive\Desktop\FinancialForecasting>
```